

Even-Degree Counterexamples to Conjecture 00000000095

GitHub: gaochengzhecpu

Abstract

For the fixed prime $p = 2$, every even degree $d \geq 2$ gives a counterexample to the assertion that $x^d - x - p$ has Galois group S_d over $\mathbb{Q}$. The rational root -1 forces the actual Galois group to have order at most $(d-1)! < d!$. The proof rules out even an abstract group isomorphism with S_d , and the counterexamples have arbitrarily large degree.

The statement

The bilingual source asserts that for every fixed prime p , as $d \rightarrow \infty$, the polynomial $x^d - x - p$ has Galois group S_d over $\mathbb{Q}$. Its usual eventual interpretation is

$$\forall p \text{ prime } \exists D \forall d \geq D, \quad \text{Gal}(x^d - x - p/\mathbb{Q}) \cong S_d.$$

Here the Galois group of a polynomial is the automorphism group over $\mathbb{Q}$ of its splitting field. Both source versions include $p = 2$ and impose no parity restriction on d .

Theorem 1. *For every even integer $d \geq 2$, let $f_d(x) = x^d - x - 2 \in \mathbb{Q}[x]$. Then*

$$|\text{Gal}(f_d/\mathbb{Q})| \leq (d-1)! < d!.$$

Consequently $\text{Gal}(f_d/\mathbb{Q})$ is not isomorphic to S_d .

Proof. Since d is even,

$$f_d(-1) = (-1)^d - (-1) - 2 = 1 + 1 - 2 = 0.$$

The factor theorem gives $f_d = (x+1)q_d$, where $q_d \in \mathbb{Q}[x]$ has degree $d-1$. A splitting field of q_d already contains the additional root -1 , so it is also a splitting field of f_d . Thus these polynomials have isomorphic Galois groups.

Let R be the set of distinct roots of q_d in its splitting field. Every automorphism over $\mathbb{Q}$ permutes R . This action is faithful: an automorphism fixing all roots fixes the field they generate over $\mathbb{Q}$, which is the entire splitting field. It therefore embeds the Galois group into the permutation group of R . Since $|R| \leq \deg q_d = d-1$,

$$|\text{Gal}(f_d/\mathbb{Q})| = |\text{Gal}(q_d/\mathbb{Q})| \leq |R|! \leq (d-1)!.$$

Finally $d! = d(d-1)! > (d-1)!$ for $d \geq 2$. Finite isomorphic groups have the same cardinality, and $|S_d| = d!$, so the claimed isomorphism cannot exist. $\square$

For every degree threshold $D \geq 0$, choose the even degree $d = 2(D+1)$. Then $d \geq D$, $d \geq 2$, and Theorem 1 applies at the same fixed prime 2. Hence no eventual threshold exists for that prime, disproving the conjecture. The failure occurs at every positive even degree, so it also precludes interpreting the claim as holding for a set of degrees of density one. We make no assertion here about the odd degrees, or about a modified statement restricted to odd primes.

Formal verification

The Lean project uses Mathlib’s actual polynomial type $\mathbb{Q}[X]$ and `Polynomial.Gal`, defined as the automorphism group of the polynomial’s actual splitting field. It does not replace the Galois group by a declared finite group or infer the abstract-group conclusion solely from reducibility.

The proof in `lean/Main.lean` verifies the following correspondence:

- The polynomial `trinomial` is $X^d - X - C(p)$, with verified degree d for $d \geq 2$. Its value at -1 is proved to be zero for every even d when $p = 2$.
- The general bound on a polynomial’s Galois group uses Mathlib’s faithful root action. The number of distinct roots is bounded by the degree, giving the factorial bound without any unproved irreducibility assumption.
- The linear-factor step uses Mathlib’s injective restriction map from the Galois group of a product to the product of the two factor groups. The group of $X - C(r)$ is trivial for rational r . This proves the required inequality for the product directly; a separate equality of splitting fields is not needed by the formal proof.
- `even_degree_gal_card_bound` gives the upper bound $(d-1)!$. The next theorem excludes an actual multiplicative equivalence with `Equiv.Perm (Fin d)`, so the conclusion also excludes an abstract isomorphism.
- `arbitrarily_large_counterexamples` constructs the degree $2(D+1)$ for every D . The theorem `conjecture95_false` negates the quantified eventual statement for every fixed prime.

All these are generic theorems. The optional Python script checks the root, exact polynomial factorization, and factorial arithmetic for six sample even degrees; it does not compute exact Galois groups or justify the infinite result by finite experimentation. The density-one observation above is an elementary consequence of failure at every even degree; the Lean conclusion is the explicit unbounded-family result and the negation of the eventual assertion.

Source and reproduction

The original statement is preserved byte-for-byte in `SOURCE.md`. Its repository path is

`conjectures/00000000095.md`.

The project pins Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From `lean/`, run `lake build`, then

`lake env lean -DwarningAsError=true Main.lean`.

All dependency URLs are public Git URLs with locked revisions. The directory `verification/` contains actual clean-build logs, theorem-axiom audits, hashes, and PDF checks. The file `README.md` gives complete reproduction instructions.